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Paleomagnetism of 2.44 Ga mafic dykes in Russian Karelia, eastern Fennoscandian Shield — implications for continental reconstructions

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Abstract

Results of a paleomagnetic study of Paleoproterozoic dykes and Archean host rocks in the eastern Fennoscandian shield are reported here. Most of the dykes are related to ca 2440 Ma layered intrusions and have isotopic ages between 2349 and 2476 Ma. The dykes form an extensive NW–SE and E–W trending swarm. The majority are unaltered gabbronorites which carry stable remanent magnetizations with multicomponent nature. The high temperature remanence component (D), which is thought to represent the primary 2.44 Ga magnetization of the dykes, has a paleomagnetic pole at 20°S, 278.8°E (A95=6.0°, N=11 sites). Another D-like component (D′) with shallower inclination (VGP 9.6°N, 256.2°E, eight samples), at one site from baked Archean host rock may also represent the primary remanence of the dykes. The commonest remanence component (A) records an extensive Svecofennian partial overprint and is interpreted to be ca 1.84 Ga old. It yields a paleomagnetic pole at 52.6°N, 226.8°E (A95=2.7°, N=23). The fourth component, B, has a paleopole at 53.9°N, 169.6°E (A95=6.3°, N=16) and is inferred to be ca 1.75 Ga on the basis of isotopic dating, although ages of 2.1, 1.3 and 0.3 Ga are also possible.

Baked contact tests for the 2.44 Ga dykes are negative, as shown by the occurrence of component D in Archean host rocks beyond one dyke width. It is proposed that the negative baked contact test indicates regional reheating and remagnetization of the Archean basement at ca 2.44 Ga ago due to the spatially vast occurrence of coeval 2.44 Ga mafic dykes, layered intrusions, volcanics and granites in the area. Based on this hypothesis, it is suggested that component D in the dykes may be primary and acquired at ca 2.44 Ga ago.

The similarity of pole D with poles from coeval dykes and layered intrusions elsewhere in the Karelian Province and Kola Peninsula suggest that, within the Archean domain of the Fennoscandian shield, no large scale block movements have taken place since 2.44 Ga.

At 2.44 Ga the Fennoscandian Shield was located at the latitude of ca 30°S which is different to that at 1.84 and 1.75 Ga, when Fennoscandia was located at the latitudes of ca 30°N. A continental reconstruction based on pole D and the pole from the 2.45 Ga Matachewan dykes of the Superior craton in Laurentia, places the Kola Peninsula adjacent to southern Greenland. Reconstructions based on recent geological interpretations and a reconstruction yielding pole D′, are also discussed. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Dikes; Fennoscandia; Paleomagnetism; Paleoproterozoic; Reconstruction

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1. Introduction

Paleoproterozoic mafic dyke swarms dated ca 2.4 Ga occur in many of the Archean cratonized shield areas. Correlation of these dyke swarms in terms of their isotopic age, occurrence, geochemistry and paleomagnetism, has been used to investigate Paleoproterozoic plate reconstructions and possible occurrences of supercontinents (e.g. Halls, 1995; Mertanen et al., 1996; Vuollo et al., 1995, 1996). High quality paleomagnetic data which are isotopically dated can give evidence for the relative positions of the continents. The present investigation yields new paleomagnetic data from precisely dated 2.44 Ga dyke rocks in Russian Karelia, eastern Fennoscandian Shield. The dyke rocks in Russian Karelia have proven to be magnetically stable and suitable for paleomagnetic study. Preliminary paleomagnetic results were presented in Mertanen (1995) where the effect of the ca 1.9 Ga Svecofennian overprinting on the 2.4 Ga dykes was demonstrated. In this study, new samples have been taken from Paleoproterozoic dykes and from Archean basement rocks for baked contact tests. The study forms part of a wider investigation in which isotopic ages, geochemical compositions and geophysical characters of Paleoproterozoic dyke rocks in Finland and Russia are being studied. Rock magnetic investigations are given by Mertanen et al. (1998). Geochemistry and isotopic datings will be presented by Vuollo et al. (1999a,b).

In a paleomagnetic study of 2.4 Ga Koillismaa layered intrusions in Finland (Mertanen et al., 1989) a primary 2.4 Ga old remanence was suggested although the primary nature could not be verified due to mild tilting of the intrusions and the occurrence of several superimposed remanence components. Since then, several paleomagnetic studies on 2.4 Ga layered intrusions and associated mafic dykes in the Karelian and Belomorian Provinces and Kola Peninsula have been carried out. Paleomagnetism of the large Burakovka layered intrusion in the SE Fennoscandian Shield was studied by Krasnova and Gooskova (1990) and by Khramov et al. (1997). Layered intrusions in the Kola Peninsula were recently studied by Arestova et al. (1997) and Khramov et al. (1997) and in the Belomorian Belt by Krasnova and Gooskova (1997). Paleomagnetism of mafic dykes, associated with these layered intrusions have been studied previously by Krasnova and Gooskova (1991, 1995), Mertanen and Pesonen (1990), Mertanen (1995) and Khramov et al. (1997). Although all these studies give fairly consistent results, the primary nature of a suggested 2.44 Ga old remanence component has not been conclusively established.

The main aim of the present study was to obtain new data from ca 2.44 Ga rocks. The results are used to refine the Fennoscandian paleomagnetic database and drift history of the Shield during the Paleoproterozoic. The data now allow comparison with paleomagnetic data from other 2.4 Ga formations in different blocks of the Fennoscandian Shield to determine whether horizontal movements have taken place between the blocks since 2.44 Ga. The Shield has undergone different tectonic and orogenic episodes since 2.44 Ga, the best known and studied being the Svecofennian orogeny at ca 1.8–1.9 Ga. These episodes are also recorded in the paleomagnetic component structures and they are briefly discussed here. The final goal of the study is to test Paleoproterozoic continental reconstructions which are discussed here in the light of the present paleomagnetic data and some recent reconstructions suggested for Laurentia and Fennoscandia (Heaman, 1997).

2. Geological setting

Paleoproterozoic continental rifting at ca 2.45–2.40 Ga in the Fennoscandian Shield is manifested by intrusion of voluminous mafic–ultramafic layered intrusions, mafic dykes and bimodal volcanic suites (e.g. Vuollo, 1994; Amelin et al., 1995). In Fennoscandia, the layered intrusions were emplaced in two discontinuous NW–SE trending belts, another in the Kola Province, and another in the Karelian Province in central Fennoscandia, extending from Tornio in NE Finland towards the SE to the Burakovka intrusion in Russia. The layered intrusions were emplaced in two short episodes, the older group at ca 2505–2501 Ma in the Kola Province and the younger group at ca

2449–2441 Ma in both the Karelian and Kola Provinces (Alapieti, 1982; Balashov et al., 1993; Amelin et al., 1995; Amelin and Semenov, 1996). All layered intrusions are associated with mafic dyke swarms intruded at about the same time.

The present study area (Fig. 1) in Russian Karelia belongs to the Late Archean Tavayarvi diorite–plagiogranite complex which yields U–Pb age (zircon) of 2724 ± 8 Ma (Bibikova et al., 1997). The area is located south of the Olanga layered complex where U–Pb ages (zircon and baddeleyite) of the Lukkulaisvaara, Tsipringa and Kivakka intrusions range from 2441 to 2445 Ma (Balashov et al., 1993; Amelin et al., 1995). Intrusion of layered complexes and related dykes was accompanied by extrusion of extensive mafic and felsic volcanism. The Koillismaa layered complex in Finland and the Olanga complex in Russia (Fig. 1) are mainly located at the boundary between Paleoproterozoic supracrustal rocks and 2.9–2.7 Ga Archean granitic gneiss basement. Different models for the origin of the rift-related igneous activity range from mantle plumes to mantle hotspots [see discussion in Amelin et al. (1995), Ernst and Buchan (1997), Sorjonen-Ward (1997)].

In the present study area (Fig. 1) the dykes have a general NW–SE or E–W trend and subvertical dips. Widths of the dykes exhibit large variation, ranging from 6 cm to 200 m. However, no general pattern between the trend and width of the dyke was observed. Compositions of the studied dykes range from gabbronorites and calc alkaline porphyritic dykes to tholeiitic and Fe-tholeiitic dykes (Vuollo et al., 1999a).

Four dykes have been isotopically dated. A gabbronorite dyke BD in the Suoperä area (Fig. 1) has a U–Pb baddeleyite age of 2446 ± 5 Ma (Vuollo et al., 1995, 1996, 1999a,b) and a Sm–Nd model age of 2421 ± 30 Ma, with ϵ-value of −2.5 ± 0.3 (Vuollo et al., 1999a). Gabbronorite dyke XD1 (Figs. 1 and 11) has a Sm–Nd age of 2422 ± 35 Ma ($\epsilon = -1.4 \pm 0.4$). Fe-tholeiitic dyke XD3 has a Sm–

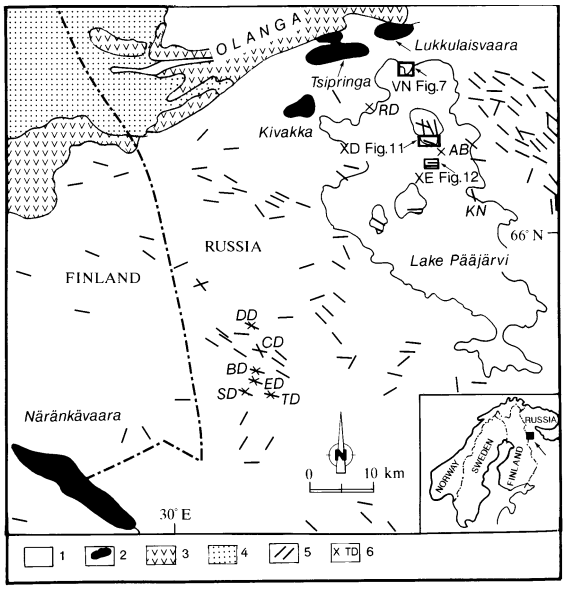


Fig. 1. Simplified geological map of the study area in Russian Karelia, compiled from Turchenko (1992), V.S. Stepanov and A.I. Slabunov (personal communication, 1994). 1, Archean granite gneiss basement; 2, layered intrusions (2.45–2.35 Ga); 3, Sumi-Sariolan (2.5–2.3 Ga) volcanic and sedimentary rocks; 4, Jatulian (2.2–2.0 Ga) volcanic and sedimentary rocks; 5, mafic dykes (ca 2.4 Ga); 6, sampling site. Areas of Figs. 7, 11 and 12 are outlined.

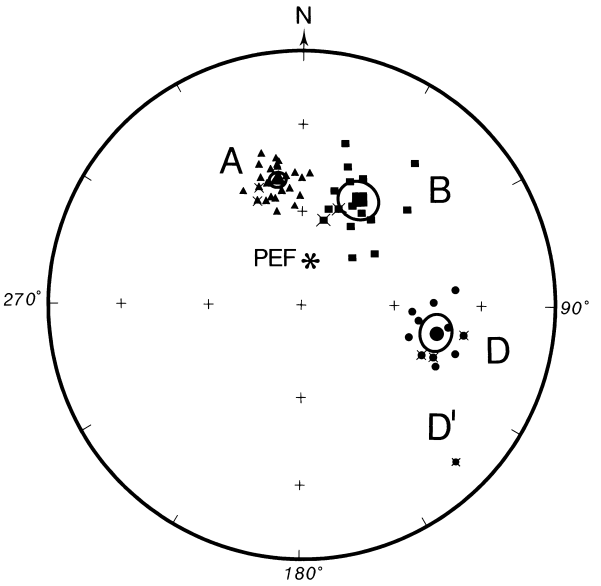


Fig. 2. Site/dyke mean directions of components A, B, D and D′. Symbols with crosses are from Archean basement, the others from dykes. Larger symbols with α95 confidence cones, denote the mean directions of components. Component D′ is from the Archean basement at site VN.

Nd age of 2476±30 Ma ($\epsilon = +1.7 \pm 0.3$) and the youngest cross cutting dyke, the Fe-tholeiitic dyke XD4, has a Sm–Nd age of 2349±30 Ma ($\epsilon = +1.0 \pm 0.3$) (Vuollo et al., 1999a).

Most of the dykes are unaltered except for dyke KN (Fig. 1), which is markedly green. Also, in the layered intrusions of the Olanga complex the primary igneous minerals and textures are generally well preserved (Amelin and Semenov, 1996). However, minor greenschist facies metamorphism is seen as alteration of olivine and pyroxene to ferromagnesian amphibole or biotite and alteration of plagioclase to saussurite. Metamorphic overprinting is also demonstrated by several isotopic dates (see Amelin and Semenov, 1996). In the Tsipringa and Lukkulaisvaara layered intrusions, metamorphic disturbance is estimated to have taken place at 1920±330 Ma on the basis of Rb–Sr data (Amelin and Semenov, 1996). These intrusions also exhibit concordant U–Pb ages on apatite and sphene of 1755 Ma (Amelin et al., 1995). Sumian felsic volcanics exhibit Rb–Sr whole rock ages from 1800–1750 Ma (Gorokhov et al., 1982). The metabasalts and metarhyolite porphyries from the Tsipringa–Panajärvi structure exhibit K–Ar system resetting at 1680–1665 Ma seen as growth of high-K metamorphic amphibole and at 1520–1490 Ma when secondary biotite and muscovite were grown in local shear zones (Buiko et al., 1995). A similar age of 1500 Ma is obtained from Rb–Sr data on the Kivakka intrusion where minor replacement of pyroxenes by biotite has taken place (Amelin and Semenov, 1996).

3. Sampling and laboratory procedure

Samples for paleomagnetic studies were taken from 36 dykes and results for 28 dykes are reported here. The discarded eight dykes showed unstable or sporadic remanence directions. A minimum of five samples was taken from each site, ca 80% as core samples, the rest as hand samples. Orientation was done with sun and magnetic compasses. Deviations in orientation directions between the two methods are typically below 5°. Two to three specimens were measured from each sample. At four locations (VN, XD, AB and CD) the dykes show cross cutting relations where either a NW–SE dyke cuts an E–W dyke, or an E–W dyke cuts a NW–SE dyke. Samples for baked contact test were taken from sites VN, XD and XE, where parts of the dykes are totally exposed and exhibit chilled margins with sharp contacts.

Magnetic cleaning methods, both alternating field (AF) and thermal, and equipment used are described in Mertanen (1995). A two-stage cleaning method, involving AF demagnetization followed by thermal damagnetization, was applied for part of the collection (see Mertanen, 1995). Samples were measured with Schonstedt spinner magnetometer and a SQUID magnetometer. Remanence components were visually inspected with vector diagrams (Zijderveld, 1967) and directions of isolated components calculated by principal component analysis (Kirschvink, 1980; Leino, 1991). Mean remanence directions and pole positions were calculated using Fisher statistics (Fisher, 1953). In addition to intensity of remanence (NRM), susceptibility and density were measured from each specimen. Petrophysical and rock magnetic investigations, comprising hysteresis measurements, thermomagnetic curves and AMS studies, are reported in Mertanen et al. (1998b). At least two polished thin sections were studied from each dyke and their host rocks.

4. Paleomagnetic results

The intensity of remanence (NRM) of the dykes ranges typically between 1 and 9 A m^{-1}, the susceptibilities between 12 000 and 70 000 (10^{-6} SI), and Q-values between 0.9 and 4.5. No clear distinction between different dykes can be made but all dykes form a fairly coherent group. In those few cases where the dykes are clearly altered, the intensities of remanence and susceptibilities are an order of magnitude lower than in other dykes.

Multicomponent analysis reveals dominantly three components, a NW intermediate-down directed A component (Table 1), a NE directed intermediate-down B component (Table 2) and an ESE intermediate-down directed D component (Table 3). In addition, the Archean host rock at site VN and one sample of the tholeiitic dyke at

Table 1
Remanence component A (Lat = 66.0°N, Long = 30.7°E)[a]

Dyke/site	B/N/n	D (°)	I (°)	$\alpha 95$	k	Plat	Plong	dp	dm
BD	*9/12	341.2	54.1	6.1	73	55.1	238.5	5.7	8.3
CD1 and CD2	*7/13	344.3	53.4	7.6	64	56.8	234.4	7.4	10.6
DD	*6/17	359.4	54.4	7.9	73	59.2	211.0	7.8	11.1
ED	*3/5	344.5	38.1	18.6	45	44.5	230.7	13.0	22.0
SD	*3/7	344.2	48.6	4.0	954	52.6	233.1	3.5	5.3
TD	*11/30	345.8	54.4	5.9	60	58.1	232.5	5.9	8.4
VN1	*4/8	352.6	47.4	8.1	130	52.0	221.5	6.8	10.5
VN2	*7/11	354.3	51.8	4.4	192	56.0	219.5	4.1	6.0
VN3	*3/7	350.2	52.2	10.2	149	56.1	225.8	9.6	13.9
VN4	*2/5	353.5	41.3	–	–	47.3	219.7	–	–
VN5	*2/2	356.8	46.6	–	–	51.6	215.5	–	–
VN6[b]	1/*1	355.5	57.4	–	–	61.7	218.4	–	–
XD-A	*5/13/24	339.6	48.9	8.2	88	51.5	240.0	7.2	10.8
XD1	*6/7	350.2	52.1	9.3	53	56.0	225.7	8.8	12.8
XD2	*3/5	359.5	48.5	11.8	110	53.2	211.6	10.2	15.6
XD3	*6/14	349.5	43.1	9.8	48	48.3	225.3	7.5	12.2
XD4	*4/7	3.5	46.8	5.4	288	51.8	206.0	4.5	7.0
XE-A	*3/4	337.1	52.9	12.4	100	54.6	245.2	11.8	17.1
XE1	*4/5	349.5	52.0	8.1	130	55.9	227.0	7.7	11.1
XE2[b]	1/*1	345.0	58.5	–	–	61.8	236.1	–	–
AB1	*3/3	350.3	40.8	28.3	20	46.8	224.0	20.8	34.4
AB2	*8/12	347.4	44.2	6.9	65	49.0	228.4	5.4	8.7
RD1	*3/4	343.1	41.6	10.2	146	46.4	233.5	7.7	12.5
RD2[b]	*1/1	333.2	47.4	–	–	48.9	247.8	–	–
RD3	*2/3	350.0	43.8	–	–	48.9	224.6	–	–
KN	*4/6	341.7	46.1	6.7	187	49.8	236.7	5.5	8.6
Mean	*23/116/211	349.1	48.2	2.5	148	52.6	226.6	$A95 = 2.7$	$K = 127$

[a] Note: in XD-A and XE-A, A denotes Archean, B/N/n is the number of dykes or sites/samples/specimens used in the statistical calculations. D and I are the mean declination and inclination respectively; $\alpha 95$ is the radius of the circle of 95% confidence; k is Fisher's (1953) precision parameter; Plat and Plong are the paleolatitude and paleolongitude for the Virtual Geomagnetic Poles; dp and dm are the semi-axes of the oval of 95% confidence; $A95$ is the radius of the circle of 95% confidence of the mean poles.

[b] Not included in mean direction.

* Denotes the statistical level used in mean calculations.

site AB carry an ESE shallow-down directed D′ component (Table 4). Typically, each specimen carries two components. Fig. 2 shows the mean directions of different components. Virtually identical behaviour also characterises NW and WNW-trending dykes in the Taivalkoski and Suomussalmi areas of eastern Finland (H. Halls, unpublished data).

The most common component, isolated in all dykes and in most of the samples from the Archean basement, is component A (Table 1). Component A was best separated by thermal demagnetization in a temperature range of ca 400–560°C. Component B has an intermediate direction between components A and D and typically the direction is close to that of component A. Therefore, in many cases it was impossible to make a clear distinction between A and B components. In Mertanen (1995), remanence directions with declinations up to 30°, were calculated together with the A component, although it was suspected that another component superimposed on the A component might exist. Here, a remanence component is defined to be component B (Table 2) if the declination is >10 and component A, if the declination is <10. When A and B occur together, B has lower unblocking temperatures than A. When superimposed with component D, B has lower

Table 2
Remanence component B[a]

	B/N/n	D (°)	I (°)	$\alpha 95$	k	Plat	Plong	dp	dm
BD[b]	1/*1	12.8	56.0	–	–	59.8	189.4	–	–
DD	*4/9	46.9	67.9	6.9	177	62.2	129.3	9.7	11.6
ED	*4/7	15.2	34.6	16.1	34	42.3	190.6	10.6	18.5
TD	*4/6	54.9	61.4	9.4	97	52.2	130.4	11.1	14.4
VN-A	*3/5	20.9	56.8	21.4	34	58.7	177.8	22.5	31.0
VN1	*5/7	18.5	42.3	15.8	24	46.7	186	12.0	19.5
VN2	*3/5	15.5	57.9	4.7	703	60.9	185.5	5.0	6.8
VN5	*2/2	38.8	29.6	–	–	33.6	164.6	–	–
XD-A	*2/2	14.3	61.9	–	–	65.6	185.0	–	–
XD1	*3/6	21.4	46.9	12.3	101	49.7	181.0	10.3	15.9
XD2	*7/15	16.0	51.5	5.0	148	54.6	187.0	4.6	6.8
XD3[b]	1/*2	20.1	48.7	–	–	51.4	182.3	–	–
XD4[b]	1/*1	39.6	54.8	–	–	51.5	154.2	–	–
XE1	*2/2	48.1	43.5	–	–	39.5	150.4	–	–
XE2	*5/5	39.1	54.5	7.4	109	51.4	155.1	7.3	10.4
AB1[b]	1/*1	36.4	57.8	–	–	55.4	156.0	–	–
AB2	*4/4	27.1	54.0	17.6	28	54.6	170.6	17.3	24.6
RD1	*3/7	33.1	54.5	24.9	26	53.4	162.4	24.8	35.2
RD2	*3/3	31.9	60.2	16.8	55	59.3	159.5	19.3	25.5
KN	*3/5	26.0	44.2	11.0	126	46.7	176.1	8.7	13.8
Mean	*16/57/135	28.5	51.4	6.3	35	53.9	169.6	$A95=6.3$	$K=33$

[a] Note: for explanations see Table 1.
[b] Not included in mean direction.
* Denotes the statistical level used in mean calculations.

unblocking temperatures, similar to unblocking temperatures of component A.

Coercivities of A and/or B and D components are largely or totally overlapping, yielding curved vector diagrams (Zijderveld, 1967) (see Fig. 3). Therefore, component D (Table 3) was isolated only by thermal demagnetization, typically in a very narrow unblocking temperature range of ca 560–585°C. Intensity decay curves for samples yielding a large D component are square shouldered (Fig. 4), while samples yielding only components A and/or B (Fig. 5) or a small D component, show more distributed demagnetization decay curves (Fig. 6).

In the following, results are presented from those sites where three or four dykes occur and where baked contact tests were carried out (sites VN, XD and XE). In Mertanen (1995) the component D was interpreted as a primary ca 2.44 Ga old magnetization and component A as a Svecofennian overprint acquired ca 1.9–1.84 Ga ago. Descriptions of demagnetization behaviour from individual dykes at site XD and at Suoperä (dykes BD, CD, DD, SD and TD) as well as preliminary interpretations of the origins of different components are discussed in Mertanen (1995).

4.1. Varisniemi VN

In Varisniemi several dykes occur which partly cross-cut each other (Fig. 7). Samples were taken at two locations, in the northern [Fig. 7(a)] and southern [Fig. 7(b)] parts of the peninsula. Gabbronorite dyke VN1 shows identical demagnetization behaviour with the dated gabbronorite dyke BD [2446 Ma, Vuollo et al. (1995, 1996, 1999a,b)] and the gabbronorite dyke XD1 [2442 Ma, Vuollo et al. (1999a)]. A low T_{UB} component A or B and a high T_{UB} component D were isolated by thermal demagnetization (Fig. 3). In some samples components A and D are superim-

Table 3
Remanence component D[a]

	B/N/n	D (°)	I (°)	$\alpha 95$	k	Plat	Plong	dp	dm
BD	1/*9/13	106.8	53.8	4.0	168	−24.5	270.5	3.9	5.6
CD2[b]	1/1/*1	89.3	61.6	–	–	−38.5	279.9	–	–
SD	1/*4/9	93.7	54.2	15.2	381	−29.9	281.2	15.0	21.4
VN-A	*8/9/14	112.4	43.7	7.1	62	−14.8	270.3	5.5	8.9
VN1	1/*4/5	116.2	40.9	9.9	87	−11.5	268.1	7.3	12.0
VN2[b]	1/1/*1	109.5	33.3	–	–	–9.1	275.7	–	–
VN7[b]	1/*1	107.5	44.5	–	–	−17.1	274.5	–	–
XD-A	*6/16/27	113.1	47.7	8.7	60	−17.5	268.6	7.4	11.4
XD1	1/*5/9	108.3	36.8	10.5	54	−11.8	276.4	7.2	12.3
XD2	1/*5/11	84.7	39.2	10.2	57	−22.3	296.3	7.3	12.2
XE-A	1/5/*5	100.7	35.5	16.0	24	−13.7	283.3	10.7	18.6
XE1	1/*7/9	97.9	51.8	11.7	28	−26.3	279.9	10.9	15.9
XE2	1/*3/4	89.3	47.0	21.1	35	−25.9	289.3	17.6	27.3
XE3[b]	1/1/*1	96.0	30.0	–	–	−12.3	288.9	–	–
XE4[b]	1/1/*1	80.3	37.4	–	–	−23.0	301.0	–	–
AB2	1/*2/3	99.1	41.4	–	–	−18.1	283.0	–	–
RD[b]	1/1/*4	98.5	51.5	–	–	−25.9	279.3	–	–
Mean	*11/69/109	102.1	45.2	5.6	67.3	−19.9	278.7	$A95=6.1$	$K=58$

[a] Note: for the virtual geomagnetic poles (Plat, Plong), reversed polarity has been used. For other explanations see Table 1.
[b] Not included in mean direction.
* Denotes the statistical level used in mean calculations.

Table 4
Remanence component D′[a]

	B/S/N/n	D (°)	I (°)	$\alpha 95$	k	Plat	Plong	dp	dm
VN1-A	*4/6/11	132.9	12.6	11.9	61	9.9	258.2	6.2	12.1
VN7-A	*4/4/8	136.6	15.9	11.3	67	9.3	254.2	6.0	11.7
AB2[b]	1/*5	145.2	12.2	–	–	13.4	246.7	–	–
Mean	*2/8/10/19	134.8	14.2	–	–	9.6	256.2	–	–

[a] Note: VN1-A/VN7-A denotes the Archaean sites at the vicinity of dykes VN1 and VN7, respectively. B/S/N/n is the number of dykes/sites/samples/specimens. For the virtual geomagnetic poles (Plat, Plong), reversed polarity has been used. For other explanations, see Table 1.
[b] Not included in mean direction.
* Denotes the statistical level used in mean calculations.

posed, in others, components B and D, but components A and B do not occur together in a same specimen.

Most samples of the Fe-tholeiitic dyke VN2 [Fig. 7(a)] carry components A and B, but one sample taken at the margin of the dyke yields component D (specimen VN6-1C, Fig. 6). Intensity and demagnetization behaviour of this sample are identical to that of dyke VN1. However, dyke VN2 is more severely overprinted by components A and B than dyke VN1, probably due to compositional differences and coarser grain size of dyke VN2 which has caused the dyke to be more vulnerable to remagnetization. Because component D was isolated only in one sample in dyke VN2, it was not used in the mean calculations.

The Fe-tholeiitic dyke VN3, which is greenish in color, yields only component A. However,

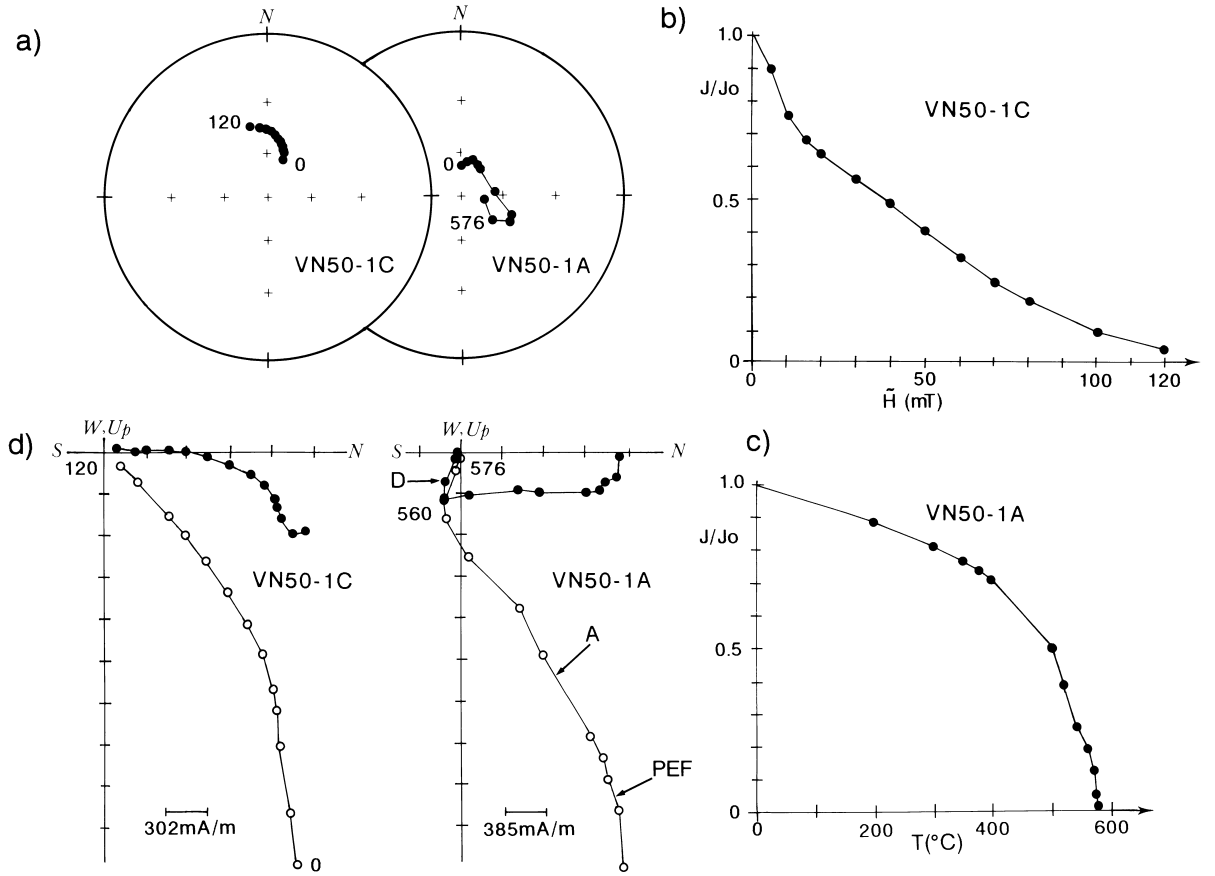


Fig. 3. Example of companion specimens showing differences in response to AF and thermal demagnetization. Specimen VN50-1C, demagnetized with AF, shows curved vector diagrams when two components are demagntized simultaneously. Specimen VN50-1A was thermally demagnetized and reveals three components, the PEF, A and D. (a) Stereographic projection of directional data upon demagnetization; (b) relative NRM intensity decay curve upon AF demagnetization; (c) NRM intensity decay curve upon heating; and (d) orthogonal demagnetization diagrams. Open (closed) symbols denote projections onto vertical (horizontal) planes. Numbers at demagnetization steps denote peak AF (mT) (VN50-1C) or temperatures (°C) (VN50-1A).

samples taken from dyke VN1 at the contact to dyke VN3 [Fig. 7(a)], reveal only the B component which could consequently be primary for dyke VN3, although the dyke itself was totally overprinted. However, because the B component was isolated also far from dyke VN3, in dykes VN1 and VN2, the baked contact test is negative and B is thus a secondary overprint.

The gabbronorite dykes VN4, VN5 and VN6 [Fig. 7(b)] carry secondary components A and B (see Tables 1 and 2). In the Fe-tholeiitic dyke VN7 two samples were taken. One is unstable and the other yields component D (Table 3).

4.1.1. Baked contact tests

Baked contact tests (Everitt and Clegg, 1962) were carried out for dykes VN1 and VN7 (see Figs. 7 and 8). For dyke VN1 [Fig. 7(a)], component D was isolated in one sample right at the contact to the dyke (sample 48). In addition to component D, this sample carries another component, called component D′ which was isolated also in two other contact samples (40 and 41) as a single component. The declination of D′ is higher and the inclination shallower than component D (Fig. 2, Table 4). The D′ component has higher unblocking temperatures and coercivities than the

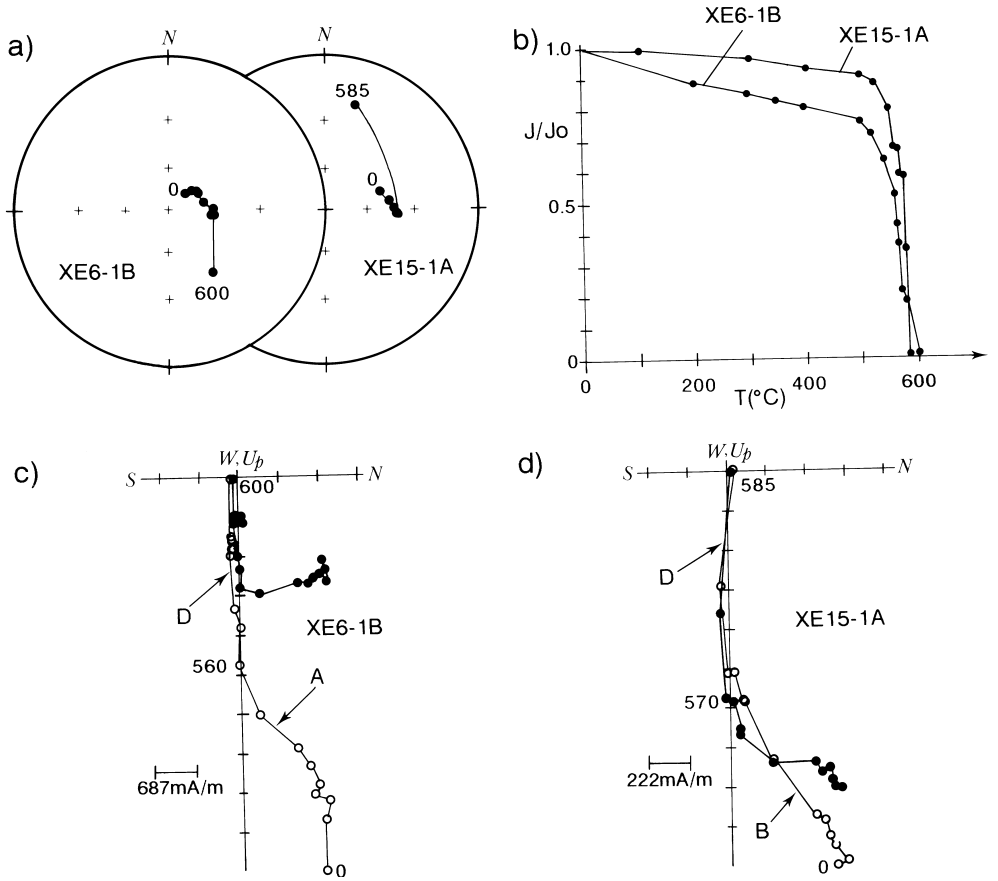


Fig. 4. Examples of thermal demagnetization behaviour for dyke XE1 (specimen XE6-1B) and dyke XE2 (specimen XE15-1A). Both dykes exhibit a relatively large D component and smaller components A and B, respectively. Symbols as in Fig. 3.

D component. AF (Fig. 8) and thermal cleaning give consistent directions.

In sample 37, taken 50 cm from the dyke, only the D′ direction was obtained. In sample 36, 1.3 m from the dyke, both D and D′ components occur in the same specimen. Two samples (50 and 51), taken >10 m from the dyke, yield a direction close to component D as the low coercivity and low T_{UB} component. Another component of almost opposite polarity but with shallower inclination was isolated in unblocking temperatures above 600°C. Component D′ was not obtained in these samples.

In the baked contact test for dyke VN7 [Fig. 7(b)], similar results with superimposed D and D′ components in samples taken at distances of 0–44 cm [samples 13,14, 15 and 29, Fig. 7(b)] from the dyke, were obtained (see Tables 3 and 4). Four samples (18–21) taken more than ten meters from the dyke, yield a steep down remanence direction, intermediate to components B and D. Results of the baked contact test are shown in Fig. 9.

The baked contact tests imply that neither the D nor D′ component is primary for dyke VN1 and VN7, because both components were isolated beyond one dyke width. Because D′ was not isolated in the VN dykes, but only in the Archean host rock close to the dykes, it is possible that D′ is not a simple thermal overprint. The generally high unblocking temperature of D' rules out viscous overprinting. It is possible that D′ is a chemical remanence (CRM) related to hydrothermal

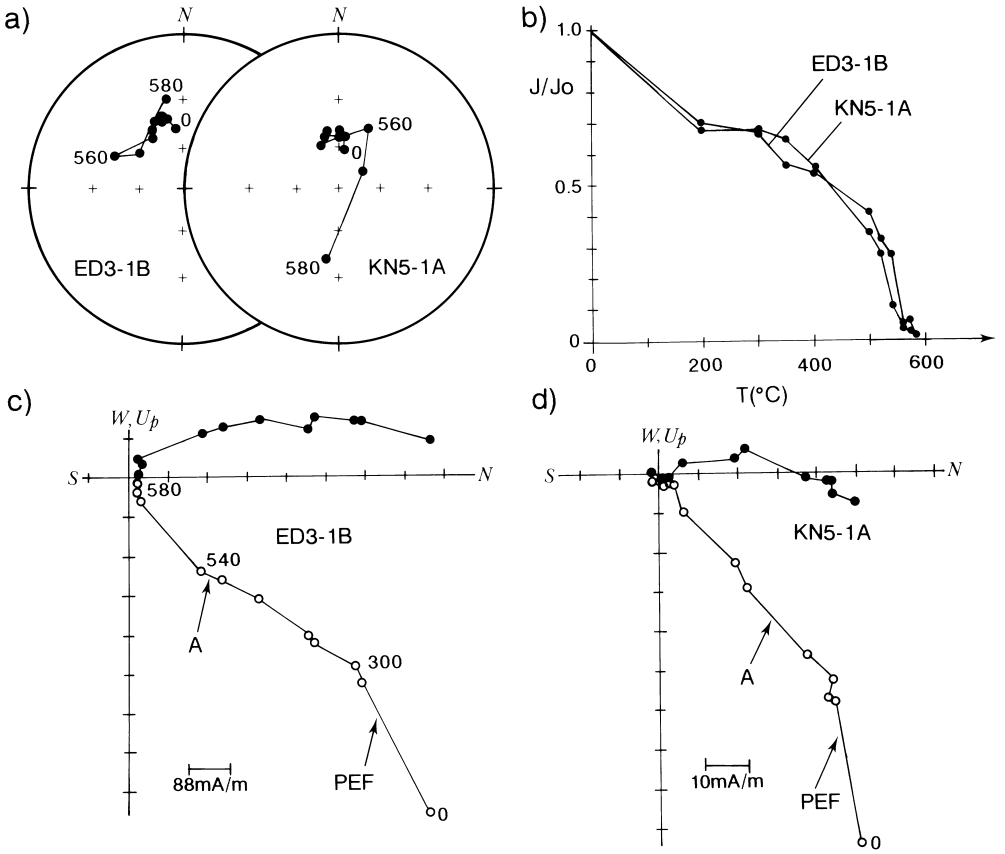


Fig. 5. Examples of thermal demagnetization behaviour for dykes ED and KN that yield component A, but not component D. Note the more distributed unblocking temperatures than in dykes XE1 and XE2 of Fig. 5. Symbols as in Fig. 3.

alteration of the host rock originating from the later heating by the dyke. Strong hydrothermal alteration of the host granodiorite, at least at the contact of dyke VN1 (sample VN41 exhibiting only component D′), is seen as totally corroded clinopyroxene which is mainly altered to hornblende and as strongly greenish outer parts of the primary brown hornblende grains. In contrast, samples VN36 and VN51 taken beyond one dyke width [Fig. 7(a)] and which carry components D and D′, are fresh and unaltered.

Similar components D and D′ were isolated in one sample in the tholeiitic dyke AB2 (Fig. 10, Tables 3 and 4), where the relative coercivities and T_{UB}s correspond to those at site VN. This is the only case observed, where component D′ is isolated in a dyke rock. Therefore, another explanation for the occurrence of component D′ also in the Archean basement at site VN, is that component D′ represents the primary remanence of the dykes. This interpretation is supported by the observation that Archean host rocks at site VN do not carry the A or B components, although these are the dominant remanence components in the dykes. It is possible that the host rocks were not remagnetized, but were able to retain their primary remanence direction, owing to the anhydrous mineral composition formed in the Late Archean high grade metamorphism (Luukkonen, 1992). In this respect, the D′ component can be considered as one candidate for the 2.44 Ga magnetization (see later).

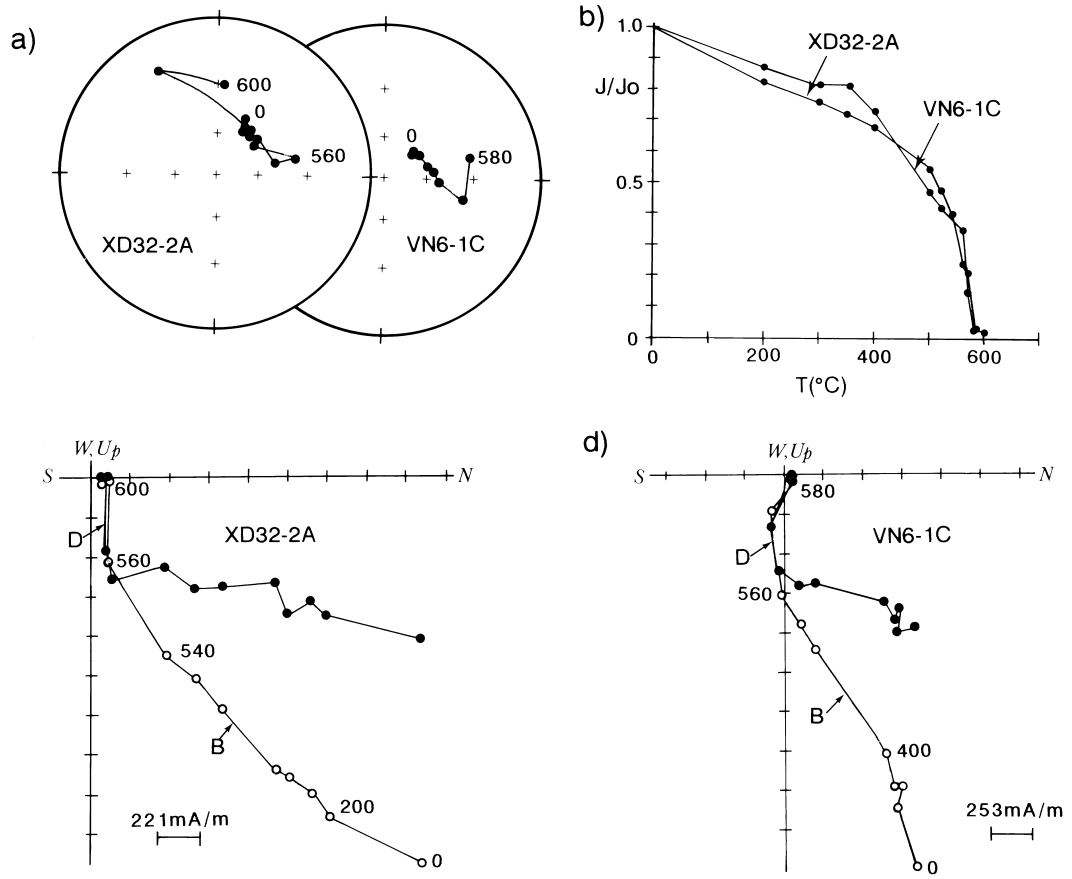


Fig. 6. Examples of thermal demagnetization behaviours of gabbronorite dykes XD1 (specimen XD32-2a) and VN2 (specimen VN6-1C) that carry components B and D. Symbols as in Fig. 3.

4.2. Lupzenga, XD

Paleomagnetic results from four cross-cutting dykes at site XD on the island of Lupzenga (Fig. 11) were given in Mertanen (1995). Since then, the dykes have been dated (Vuollo et al., 1999a,b) and new samples taken from the Archean host rocks for a baked contact test. In the gabbronorites (XD1) and in the geochemically similar, coeval dykes XD2 which occur both as thinner dykes cross cutting the gabbronorite and as a broad dyke at the western side of the gabbronorite (Fig. 11), the D component was isolated and interpreted as a primary remanence ca 2.4 Ga in age. In two other dykes, (XD3) 'Older Jatulian diabase' and (XD4) 'Younger Jatulian diabase' (Mertanen, 1995) which cross cut the gabbronorites, no D component was revealed and therefore it was suggested that the dykes are younger than the gabbronorites, probably Jatulian (ca 2.2–2.0 Ga), in age. However, new Sm–Nd dating (Vuollo et al., 1999a) suggests that, within error limits, the Fe-tholeiitic dyke XD3 (2476 ± 30 Ma) is similar in age to the gabbronorite dyke XD1 (2422 ± 35 Ma). Furthermore, the Fe-tholeiitic dyke XD4 also has a high Sm–Nd age of 2349 ± 30 Ma. This explains the apparent discrepancy in baked contact results close to the margin of the 'Older Jatulian diabase' where the D component was isolated although only a Jatulian direction was expected.

The gabbronorite dykes XD1 and XD2 typically

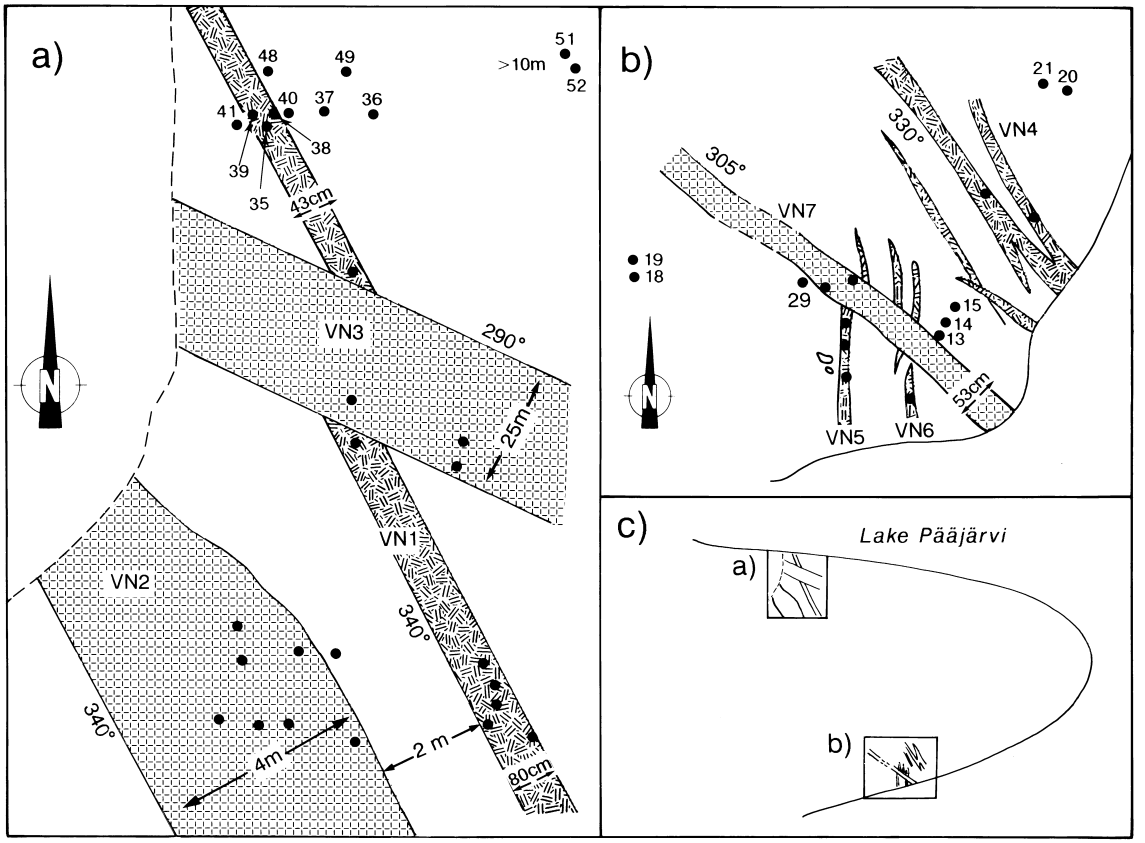


Fig. 7. Simplified geological map of site VN at Lake Pääjärvi (see Fig. 1) where baked contact tests were carried out. (a) The northern part of the peninsula, (b) the southern part of the peninsula, (c) relative locations of the two sites, not to scale.

exhibit three components, A, B and D (see Tables 1–3) which were isolated by thermal demagnetization. In both rock types component D is isolated in high T_{UB}s while A and B occur in lower temperatures. Dykes XD3 and XD4 are totally overprinted by components A and B.

4.2.1. Baked contact test

Samples (Nos 22, 65–81) for baked contact test were taken from the Archean pyroxene bearing granodiorite on both sides of the complex of gabbronorite and diabase, altogether ca 43 m wide (Fig. 11). The baked contact test on component D is negative (Table 3, Fig. 9). Unlike at Varisniemi (site VN), a consistent D component is obtained systematically from the contact margin up to the distance beyond one dyke width. Component D′ does not occur. Instead, component A is isolated in all samples at lower temperatures (Table 1).

4.3. Site XE

At site XE, two E–W trending gabbronorite dykes, XE1 and XE2 (Fig. 12), transect the granitic Archean basement. XE1 is compositionally a porphyritic dyke of calc-alkaline affinity with a width of ca 6 m. XE2 is a tholeiitic dyke with a width of ca 16 m. The southern contact zone of dyke XE1 is glassy while the northern contact zone is tectonized. Dykes XE1 and XE2 are ca 100 m apart. In addition to dykes XE1 and XE2, two small north-northeast trending gabbronorite dykes XE3 and XE4 occur at the shore of Lake

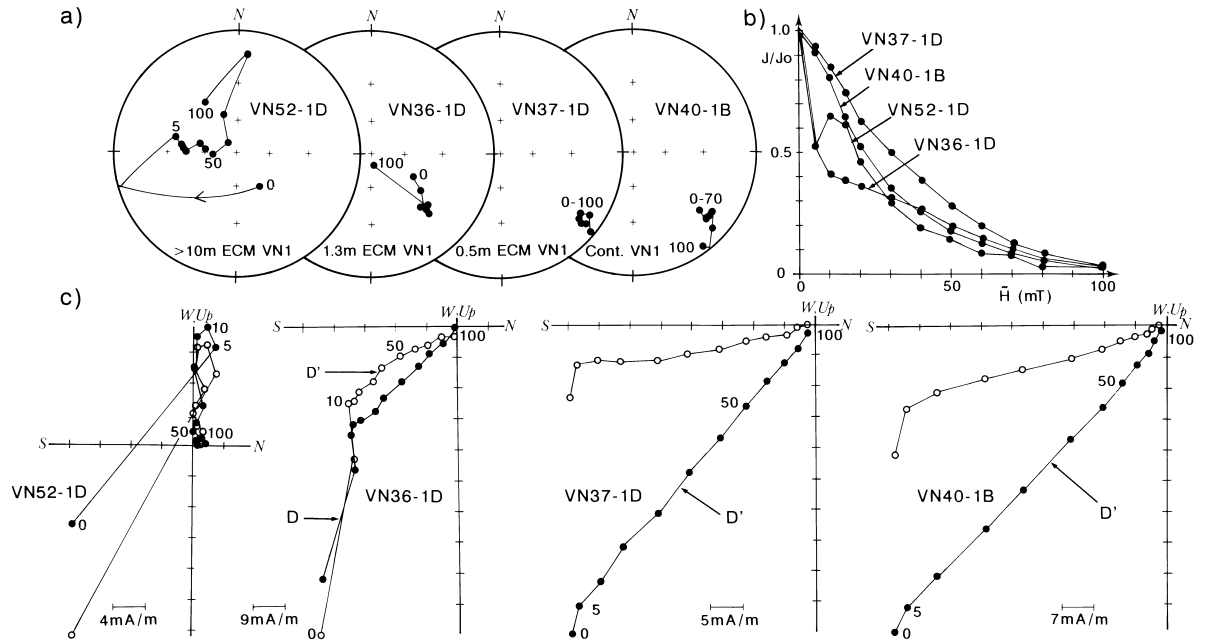


Fig. 8. AF demagnetization behaviour of the Archean rocks taken at varying distances from dyke VN1. In (a) the distance of the sample from the east contact margin (ECM) is indicated. Other symbols as in Fig. 3.

Pääjärvi. Widths of dykes XE3 and XE4 are 10 and 5 cm, respectively.

Dykes XE1 and XE2 give consistent paleomagnetic results (see Fig. 4) similar to those obtained from, for example, dykes VN2 and XD1 showing a high temperature component D (Table 3) and lower temperature components A or B (Tables 1 and 2). Dykes XE3 and XE4 have exceptionally high remanence intensities of ca 10 A m^{-1}. In these samples only a stable and hard single component D was isolated in high coercivities. Results of dykes XE3 and XE4 are not included in the mean D direction, because the direction was obtained only in one sample (Table 3).

4.3.1. Baked contact test

A baked contact test was carried out on dyke XE1. Granite samples taken from the southern contact margin and from 1.7 m to the dyke give component D. Samples taken ca 20, 35 and 40 m, thus far beyond one dyke width from dyke XE1, also give a consistent D direction (Table 3, Fig. 9). Thus, the baked contact test is negative for dyke XE1. In samples taken close to contact of dyke XE1 no clear A or B components were isolated, while in the samples far from the contact, also component A occurs (Table 1).

5. Discussion

5.1. Origin of component D

On the basis of the above results a baked contact test is negative for component D. Component D has previously been interpreted to be of primary origin and ca 2.44 Ga in age (e.g. Mertanen et al., 1989; Mertanen, 1995). In the following, the implications of the negative baked contact test on the origin of component D are discussed.

The negative baked contact test implies that component D may be a younger secondary overprint of unknown age both in the dykes and in the Archean country rocks. Recent Russian studies (Damm et al., 1997; Pisarevsky and Sokolov, 1999) support the secondary nature of component D. In southern Russian Karelia, Damm et al. (1997)

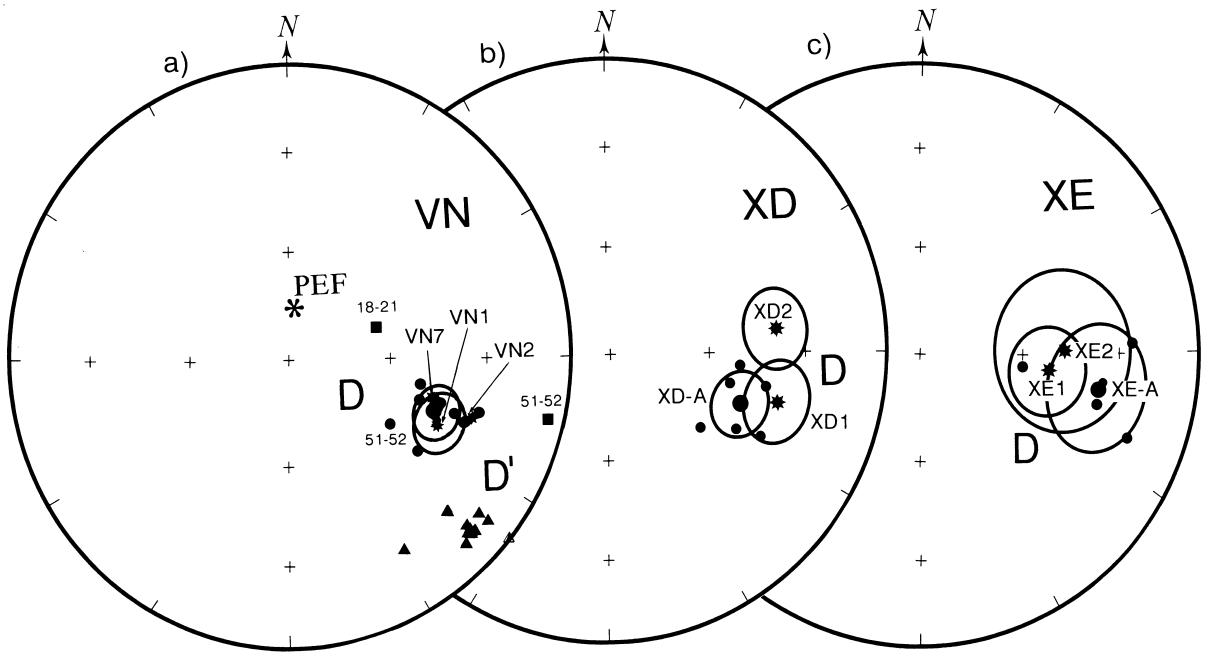


Fig. 9. Results of baked contact tests from sites (a) VN, (b) XD and (c) XE. Circles denote component *D* isolated in the Archean host rocks, stars denote the mean *D* directions of the dykes. Smaller symbols are sample mean directions, larger symbols with $\alpha 95$ confidence cones, the mean *D* directions ($\alpha 95$s are not shown for VN2 and VN7, see Table 3). In (a) triangles denote component D′ and squares mean directions from the Archean host rocks taken far beyond one dyke width from the contact. Closed (open) symbols denote projections onto lower (upper) hemispheres.

have isolated a high to moderate temperature dual polarity D-like component (their component C), probably carried by hematite, in ca 1800 Ma Shoksha quartzites and underlying basalts. In addition, they isolated a similar but soft, low temperature remanence component in the 1770 Ma old Rybreka sill which intrudes the Shoksha quartzites and basalts. In both cases the authors regard this D-like component to be of secondary origin, and <1500 Ma.

Pisarevsky and Sokolov (1999) isolated a D-like component (see Table 6 and Fig. 13), Component I, in the Konchozero peridotite sill in south-western Karelia. The sill is dated at 1974 ± 27 Ma (Sm–Nd age). Based on tentative positive baked contact test, Component I is regarded as primary. Although the baked contact test is inconclusive and therefore the age of the magnetization is not reliably defined, the occurrence of a D-like component in the sill confirms a maximum age of 1974 Ma.

Mertanen and Pesonen (1994) isolated a D-like component in the Tsuomasvarri ultramafic intrusion which on the basis of contact relations, is younger than the surrounding gabbro-diorite of 1933 Ma. However, Mertanen (1995) interpreted the D-like component in this intrusion as a Vendian remagnetization related to the emplacement of 546–580 Ma Vendian dykes close to the area (Torsvik and Meert, 1995).

Despite a negative baked contact test and occurrence of a D-like component in younger Paleoproterozoic rocks, it is suggested, however, that the component D of the present study represents a magnetization acquired at ca 2.44 Ga ago. In the Pääjärvi dykes, D is isolated in the highest unblocking temperatures, and is best obtained in the dated dyke [dyke BD, age 2446 ± 5 Ma, Vuollo et al. (1995, 1996, 1999a,b)] which is well preserved. Based on this evidence and on the general occurrence of a D-like remanence in all other 2.44 Ga formations in Fennoscandia, it is sug-

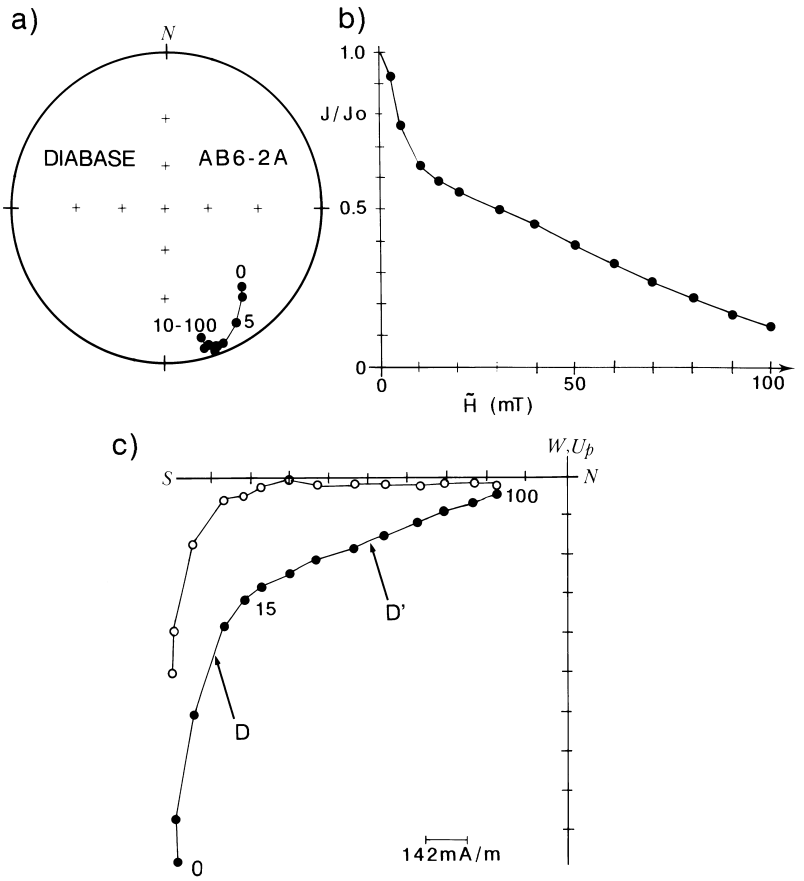


Fig. 10. AF demagnetization behaviour of a tholeiitic dyke sample from site AB. Component D is isolated in lower fields of 0–10 mT and component in higher fields of 10–100 mT. Symbols as in Fig. 3.

gested that D represents a magnetization acquired ca 2.44 Ga ago.

It is proposed that the negative baked contact test is due to remagnetization of the Archean basement of the study area ca 2.44 Ga ago. If component D were younger than component A acquired during the Svecofennian orogeny ca 1.9 Ga ago, then it would be expected to have totally reset the Svecofennian magnetization. However, in the Archean basement as well as in the dykes, the D component was systematically isolated in higher laboratory unblocking temperatures of 560–585°C compared with component A isolated in unblocking temperatures of 400–560°C. Therefore, it is concluded that component D must be older than ca 1.9 Ga [see discussion in Mertanen (1995)]. Krasnova and Gooskova (1995) have also obtained a D-like component from the Archean basement and from similar dyke rocks adjacent to the present study area further to the east. Unfortunately they do not report the distance between the dykes and the sampled country rock to evaluate possible baking effects.

If the age of component D is considered to be 2.44 Ga, it is possible that remagnetization of the Archean basement may be a local phenomena related to injection of the dykes. Local heating of the Archean basement beyond one dyke width may be due to hydrothermal supercritical solutions from the dykes which occur in a dense pattern all over in the area. At site XD (Fig. 11), all three generations of cross-cutting dykes have a Sm–Nd

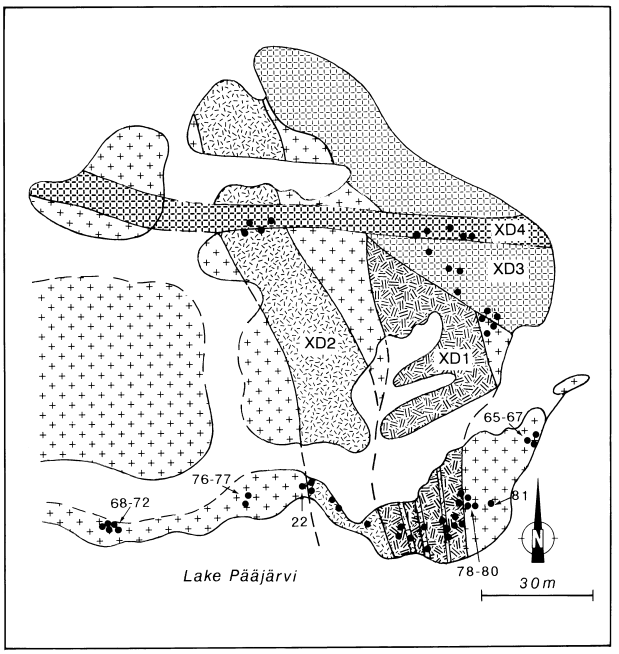


Fig. 11. Simplified geological map of site XD at Lake Pääjärvi (see Fig. 1) where baked contact test was carried out. The map was compiled after Stepanov (1994).

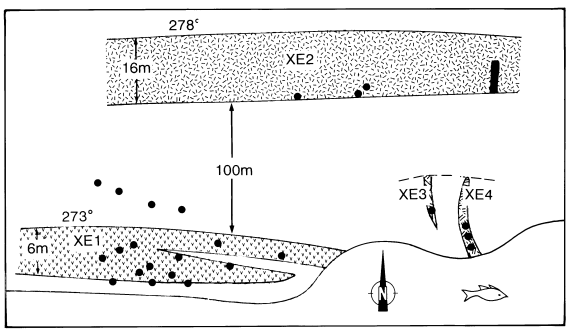


Fig. 12. Simplified geological map of site XE at Lake Pääjärvi (see Fig. 1) where the baked contact test was carried out.

model age close to 2.4 Ga. Even the youngest cross-cutting dyke is as old as 2.35 Ga (Vuollo et al., 1999a). Thus there is no age evidence for younger dyking events in the study area, unlike in eastern Finland where isotopic dates on the dykes show ages of 2.2 to 1.97 Ga (Vuollo, 1994). It is therefore possible that even those single dykes where component D was not isolated due to later remagnetization, may represent distinct pulses of the same dyking event as the dated dykes of the area. According to Oliver (1996), at midcrustal levels, fluid-related thermal activation requires channelization of the fluids, related for example, to fractures. The 2.4 Ga dyking event in the study area has been intense and it is therefore possible that initial rifting may have caused fracturing of the crust. Even if fractures were not always filled with dyke magma, they may have acted as channelways for fluid movement. However, in for example, the island of Lupzenga (site XD), where component D was invariably isolated in the whole basement area, visible fracturing was not observed and the rocks were petrographically unaltered.

According to another scenario, not only the Archean basement adjacent to dykes was heated at ca 2.4 Ga, but the entire Archean crustal block. All three areas where baked contact tests were carried out, give results in agreement. It is suggested that crustal extension at ca 2.4 Ga with upwelling magmas producing the layered intrusions and spatially dense mafic dykes may have been the heat source responsible for remagnetization. In addition, overlying volcanics, which at the present area are eroded away, may have produced

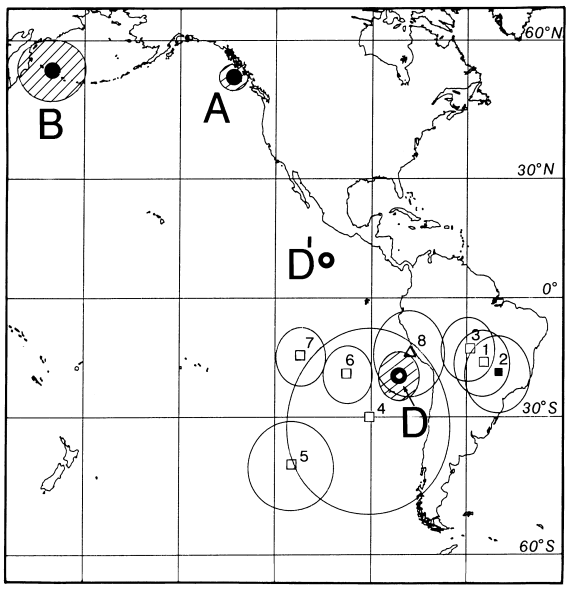


Fig. 13. Mean paleomagnetic poles for components A, B, D and D′ (circles) with shaded A95 confidence circles (except for pole D′ that was obtained only from two sites). Closed (open) symbols denote normal (reversed) polarity. Squares are poles from ca 2.45 Ga old formations in the Karelian and Kola provinces (Table 6). Triangle denotes a D-like pole from the 1974 Ga Konchozero sill (pole 8, Table 6).

heat external to the Archean basement. A U–Pb zircon age on felsic volcanics of the Lehta structure SE of the present area is 2442.0 ± 4.8 Ma (Levchenkov et al., 1990) and a Sm–Nd age on komatiitic basalt of the Vetra belt in eastern Karelia is 2448 ± 42 Ma [Puchtel et al. (1991) in Amelin et al. (1995)]. Furthermore, post-orogenic 2.44 Ga potassium granites (Luukkonen, 1992) related to the Late Archean metamorphism principally occuring in eastern Finland, are also found in the Lake Pääjärvi area. The age of the Kormanka potassium granite at the southwestern shore of Lake Pääjärvi and of a related granite porphyry further south of Lake Pääjärvi is 2449.3 ± 3.7 Ma (Levchenkov et al., 1990). All these intrusions may also have provided heat for remagnetization. The laboratory unblocking temperatures (560–585°C) of component D, imply that the temperature totally resetting the older Archean magnetization was in excess of 540°C, (Pullaiah et al., 1975). Although the Archean basement rocks and the dykes have retained their primary anhydrous mineralogy with the mafics dominated by othopyroxenes, it is probable that increased crustal heat flow was accompanied by more actively circulating groundwaters. New magnetite can be formed even at relatively low temperatures and in this sense the D magnetization could be a thermo-chemical remanent magnetization (TCRM).

Halls and Palmer (1990) have shown that dyke rocks within granulite grade areas exhibit cloudy feldspar formed by inclusions of tiny magnetite grains in the feldspar. The occurrence of cloudy feldspar in fresh dykes within granulite to upper amphibolite grade terrains is thought to be caused by magnetite exolution as a result of slow cooling of the dykes at mid to lower crustal depths (e.g. Halls and Zhang, 1998). Subsequent uplift has then brought these dykes to the surface. In the dykes of Russian Karelia, cloudy feldspar is clearly observed in dykes yielding component D while in those dykes where only A and/or B components occur, the cloudiness of plagioclase was only faint. In two cases with cross-cutting dykes the appearance of feldspar differed markedly between the dykes. At site XD the youngest cross cutting diabase dyke, XD4 is only faintly cloudy while the older dykes, XD1, XD2 and XD3 (dated ca 2.4 Ga) are distinctly cloudy. Also, at site CD [Fig. 3 in Mertanen (1995)], the younger cross-cutting dyke CD1 exhibits uncloudy feldspar while the older gabbronorite dyke, CD2 exhibits very cloudy feldspar. In dyke TD, where component D was not isolated and which was therefore regarded as younger than other gabbronorites, the plagioclase is very cloudy. In that sense dyke TD could also represent the same generation as the other gabbronorites with D component.

The occurrence of cloudy feldspar suggests that component D could be of thermochemical origin and acquired at deep crustal levels shortly after emplacement of the dykes. The remanence, which is thus secondary in the Archean host rock, was blocked as a result of uplift of the crust, probably in exess of 5–10 km (see Halls and Palmer, 1990; Mertanen et al., 1996). Such a hypothesis is presented for 2.45 Ga Matachewan dykes in Canada (Halls et al., 1994; Halls and Zhang, 1998). In the Pääjärvi area the occurrence of cloudy feldspar is

Table 5
Mean palaeomagnetic results of the dykes and host rocks[a]

Comp.	B/N	D'' (°)	I'' (°)	Plat (°N)	Plong (°E)	Pol	$A95$ (°)	K	λ_{ref} (°)
A	*22/116	347.1	46.3	52.6	226.7	N	2.8	126	28
B	*16/57	25.3	50.9	53.9	169.6	N	6.3	33	32
D	*11/65	98.1	44.4	−19.9	278.7	R	6.1	58	−26
D′	1/*8[b]	131.8	15.3	9.6	256.2	R	6.7	69	−8

[a] Note: Comp. is remanence component, B/N, number of sites/samples used for mean calculations, D'' and I'' are the declination and inclination calculatated for a common reference location (Kajaani, 64.1°N, 27.7°E), Plat and Plong are the latitude and longitude of the grand mean paleomagnetic pole, Pol is the polarity, N is normal, R is reversed polarity, $A95$ is the radius of the circle of 95% confidence about the mean paleopole, K is the precision parameter, λ_{ref} is the reference paleolatitude of the continent.
[b] Site VN.
* Denotes the statistical level used for mean calculation.

in agreement with gravity and geological observations of the present study area. Grishin (1992) has proposed that the Pääjärvi block forms an uplifted unit bordered by deep faults. The petrology of the Archean pyroxene-bearing host rocks also suggests that the area was uplifted from deeper granulite grade crustal levels. The present area was, however, later partially remagnetized and retrogressively metamorphosed to greenschist-amphibolite facies during the Svecofennian orogeny and therefore simple interpretations have to be taken with caution.

Isotopic age evidence for partial resetting and remagnetization of the Archean crust at ca 2.4 Ga is sparse. In Archean areas which show granulite facies or high amphibolite facies grade metamorphism, K–Ar dates on hornblende give ages of ca 2.5 Ga (Kontinen et al., 1992). In the Ilomantsi region of SE Finland, however, K–Ar dates on hornblende from the Archean basement give ages of ca 2.4 Ga (Kontinen et al., 1992). The U–Pb zircon age of a Syvä–Kokko hornblende gabbro in eastern Finland south of the present study area is 2700 Ma (Luukkonen, 1988), although the U–Pb titanite age is 2350 Ma (H. Huhma, personal communication, 1998). The above ages imply that at least in some areas, the Archean basement was isotopically reset at ca 2.4 Ga, in agreement with the interpretation of remagnetization of the Pääjärvi area.

5.1. Paleomagnetic poles

Paleomagnetic poles calculated from components A, B, D and D′ are listed in Table 5 and shown in Fig. 13. The origin and age of pole A were discussed in Mertanen (1995) where it was shown to have been acquired during the Svecofennian orogeny ca 1.84–1.9 Ga ago. It represents a widespred common overprint in the Archean–Paleoproterozoic Fennoscandian Shield. Pole A of this study is in agreement with pole A of Mertanen (1995).

Pole B (Fig. 13) represents a common remanence direction in the Fennoscandian Shield. In most studies, a B-like pole is interpreted to represent a secondary, 1.75 Ga old overprint related to the waning stage of the Svecofennian orogeny (e.g. Mertanen et al., 1989; Mertanen and Pesonen, 1990; Mertanen and Pesonen, 1994; Elming, 1994; Krasnova and Gooskova, 1995; Khramov et al., 1997). However, poles ca 2.1 Ga old, regarded as primary, also group with these poles (e.g. Sokolov, 1989; Khramov et al., 1997; Neuvonen et al., 1997). Furthermore, poles obtained from Proterozoic rocks younger than 1750 Ma in southern Fennoscandia have a similar position (Piper, 1979; Bylund, 1985; Mertanen and Pesonen, 1995). Mertanen and Pesonen (1995) suggest that the remanence was acquired ca 1.3 Ga ago. Even a younger, Paleozoic age, could be proposed for the B component. In a recent preliminary study by Mertanen et al. (1998a,b) a stable B-like component was isolated in a Neoproterozoic dyke in the Kola Peninsula. According to the Paleozoic APWP (Torsvik et al., 1996) the age of the remanence is 290–260 Ma. Thus, on the basis of the variety of occurrences, the age of pole B could be either ca 2.1, 1.75, 1.3 Ga or 290–260 Ma. For more detailed discussion, see Mertanen et al. (1998a,b).

Table 6
Paleomagnetic poles from Paleoproterozoic formations of the Fennoscandian Shield[a]

Pole	Formation	Pol	Plat (°N)	Plong (°E)	*A*95 (°)	M. age (Ma)	Magnetic reference	Isot. age (Ma)	Age reference
1	Main Range gabbro-diabase	N	−16.7	304.8	8.3	2400	Khramov et al. (1997)	2436±25	Khramov et al. (1997)
2	Moncha pyroxenites	R	−19.4	309.6	9.6	2300	Khramov et al. (1997)	2504±2	Amelin et al. (1995)
3	Voche-Lambina ultramafics-gneisses	M	−13.2	300.4	8.1	2300	Khramov et al. (1997)		
4	Koillismaa layered intrusion	R	−29.8	269.3	22.0	2440	Mertanen et al. (1989)	2436±5	Alapieti (1982)
5	Karelian intrusions and basement	R	−41.0	245.0	10.0	2470	Krasnova and Gooskova (1995)	2470±0.1	Amelin and Semenov (1990)
6	Burakovka layered intrusion	R	−19.5	262.6	7.2	2450	Khramov et al. (1997)	2449±1	Amelin et al. (1995)
7	Shalskiy gabbro-dolerite dyke	R	−15.0	248.0	7.4	2450	Krasnova and Gooskova (1990)		
8	Konchozero peridotite sill	R	−14.2	281.9	10.9	1974	Pisarevsky and Sokolov (1999)	1974±27	Puchtel et al. (1995)

[a] Pole numbers are used in Fig. 13. Pol is the polarity of a pole, N is normal, R is reversed, M is mixed normal and reversed polarities. Plat and Plong are the paleolatitude and paleolongitude of the pole. *A*95 is the radius of cone of 95% confidence for a pole. M. age is the interpreted magnetization age of a pole. Isot. age is the isotopic age. All isotopic ages are U–Pb dates on zircon, except for Karelian intrusions and basement (5) and Konchozero peridotite sill (8) which are Sm–Nd dates.

Much paleomagnetic data from ca 2.4 Ga old formations in Finland and Russia point to the existence of a hard D-like component. Table 6 and Fig. 13 show those poles where a D-like pole has been isolated and where the magnetization age has been regarded as ca 2.3–2.47 Ga. In addition, a pole from the Konchozero peridotite sill (pole 8) is shown. Poles 1, 2 and 3 are from the Kola Province, poles 4 and 5 from the central Karelian Province and poles 6 and 7 from the SE Karelian Province. Krasnova and Gooskova (1990) obtained a positive baked contact test for the D-like remanence of the Shalskiy gabbro-dolerite dyke (pole 7) which is genetically related with the Burakovka intrusion. In addition to these poles, other less well-defined paleomagnetic data from ca 2.44 Ga formations all point to similar results (Krasnova and Gooskova, 1990, 1991; Krasnova and Gooskova, 1995; Mertanen and Pesonen, 1990; Arestova et al., 1997; Khramov et al., 1997). Since the remanence directions from different geological provinces do not differ markedly, it seems likely that no large scale horizontal movements within the Archean Shield of Fennoscandia have taken place since 2.44 Ga. However, further paleomagnetic data from different provinces are required to verify this conclusion.

Pole D′ from the Archean basement in Varisniemi (VN) has not been commonly observed in the Fennoscandian Shield. Component D′ is obtained in rocks which do not yield the common overprints A and B, implying that the basement may have retained this (presumably older) remanent magnetization. Component D′ was detected only in the vicinity of the gabbronorite dyke, but not in the Archean basement further from the dyke. The high unblocking temperatures, above those of component D, suggest that D′ could be older than D. Therefore, it is possible that pole D′ could represent the primary remanence of the dykes and component D a slightly younger, secondary remanence related to uplift of the area at ca 2.44 Ga.

5.2. Continental reconstructions

Fig. 14 shows the orientations and paleolatitudes of the Fennoscandian Shield based on poles D, A and B of this study. Assuming that remanence component D was formed ca 2.44 Ga, the

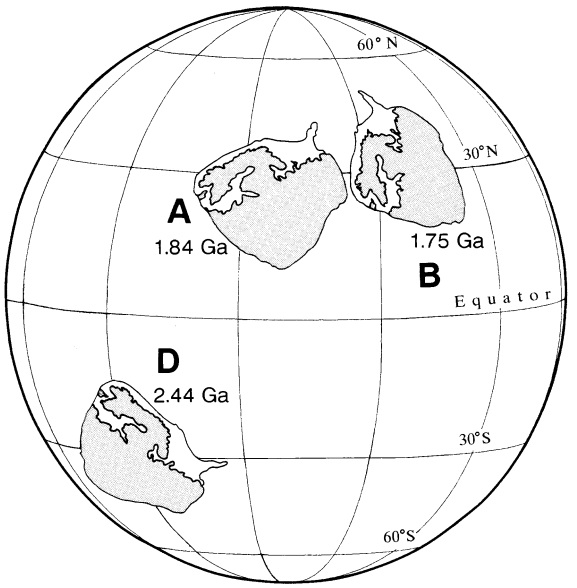


Fig. 14. Positions of the Fennoscandian Shield at ca 2.44, 1.84 and 1.75 Ga. A reversed polarity paleopole for the reconstruction at 2.44 Ga is used, for others, the normal polarity. The map was compiled by using GMAP (Torsvik et al., 1990) and MapDraw (by M. Leino, GSF).

Fennoscandian Shield was located at a latitude of 26°S at that time. During the Svecofennian orogeny ca 1.84 Ga ago, the Shield was sited at latitudes of 28°N and a latitude 32°N, probably at ca 1.75 Ga ago. Here a reversed pole [using the reversed polarity definition of Pesonen et al. (1989)] has been used for D, but due to long time gaps between poles D, A and B, the relative polarities are not known. Therefore, at 2.44 Ga the Shield may have likewise located at northern latitutes with opposite orientation.

Continental reconstructions of Laurentia and Fennoscandia at ca 2.45 Ga have recently been proposed by Heaman (1997) and discussed by Halls (1998) and Heaman (1998). Heaman (1997) suggested that the Archean Superior Province of Laurentia and the Archean Karelian Province of Fennoscandia formed parts of a supercontinent on the basis of coeval occurrence of mafic magmatism and rift-basin sedimentation at ca 2.45 Ga. His reconstruction places the southern margin of the Fennoscandian Shield adjacent to southern Superior Province so that the Matachewan dykes formed a contiguous dyke swarm with the Karelian dykes. Heaman (1997) claimed that his reconstruction was based on paleomagnetic results from the D pole of the layered intrusions of Finland (Mertanen and Pesonen, 1990) and from the pole for the Matachewan dykes of the Superior Province (Halls and Bates, 1990). Halls (1998) noted that the reconstruction is based on geological interpretations, but is not in agreement with paleomagnetic data. In his reply Heaman (1998) admitted that the reconstruction was based solely on the correlation of synchronous mafic magmatism in the two regions.

Although the magnetization age of component D cannot be definitely assigned to 2.44 Ga in this study, the reconstruction of Heaman (1997) fails to combine the Karelia and Superior Provinces correctly. In Heaman's reconstruction the paleolatitude of 45°N and orientation of the Superior Province at 2.45 Ga correspond to the present position, but on the basis of well defined paleomagnetic data from the Matachewan dykes (Halls and Bates, 1990; Bates and Halls, 1990) the Superior Province was located at a latitude of ca 10° at 2.45 Ga (either N or S) and rotated ca 60° anticlockwise/120° clockwise with respect to its present position.

To produce the reconstruction that was evidently meant by Heaman (1997), Superior is placed to its correct paleolatitude and orientation at 2.45 Ga [Fig. 15(a)], according to the Matachewan N pole [42°N, 58°E, Bates and Halls (1990)]. Fennoscandia is placed in the same relative orientation as suggested by Heaman (1997). This reconstruction for Fennoscandia is produced by using a paleopole of 20°N, 180°E (a $\pm 5°$ difference in the pole position does not significantly change the reconstruction). The reference remanence direction is D = 26°, I = −6° (calculated from the paleopole and the mean geographical position of Fennoscandia, Lat = 64.1°, Long = 27.7°). This remanence direction is close to one (D = 22°, I = −8°) currently recognized only from the Mesoproterozoic 1.56 Ga old Föglö-Sottunga diabase dykes in southern Finland [Pesonen and Neuvonen (1981); see Mertanen and Pesonen (1997)]. However, this direction has not been found in Paleoproterozoic formations

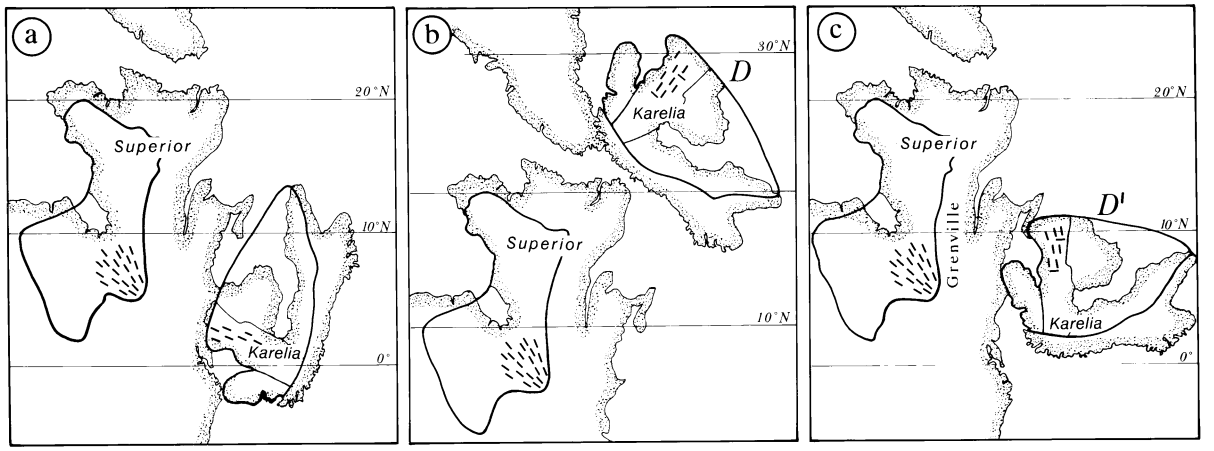


Fig. 15. Three different scenarios for paleocontinental reconstructions of Laurentia (Superior) and the Fennoscandian Shield (Karelia) at 2.45 Ga. In all figures, the Superior Province has the same paleolatitude and orientation, based on paleomagnetic pole from the Matachewan dykes (Bates and Halls, 1990). (a) Reconstruction of Heaman (1997), but modified so that Superior is in its correct paleolatitude at 2.45 Ga and Karelia is placed in the way shown in Fig. 3 of Heaman (1997). (b) Reconstruction based on pole D of this study. (c) Reconstruction based on pole D′ of this study.

(Pesonen et al., 1991; Mertanen and Pesonen, 1997). Therefore, it seems unlikely that at 2.44 Ga Superior and Karelia [Fig. 15(a)] had the configuration suggested by Heaman (1997).

Fig. 15(b) shows a reconstruction based on pole D of this study and on the pole from the 2.45 Ga Matachewan dykes (Bates and Halls, 1990). Here, a normal polarity N is used for both continents (in Fig. 14, reversed polarity for pole D was used), in order to better refer to Heaman's (1997) reconstruction. In this reconstruction based purely on paleomagnetic evidence the Kola Peninsula is adjacent to southern Greenland. The Matachewan dykes and the main part of the Karelian dykes (excluding the less often occuring E–W dykes) do not form a contiguous dyke swarm, but are oriented 60–90° with respect to each other. The reconstruction places Karelia and Kola adjacent to the Nain Province, where there is no isotopically dated evidence for 2.45 Ga dyke swarms. The main paleomagnetic difference between the Matachewan and Karelian dykes is the absence of dual polarity magnetization in the Karelian dykes which is characteristic of the Matachewan dykes. It is noted, however, that in the Kola Province, the ca 2.44 Ga gabbro-diabase dykes (see Table 6, pole 1) do show opposite polarity with the D pole of Karelia.

The Matachewan and Karelian dykes differ petrographically and geochemically. The main part of the Matachewan dykes, except the Hearst subswarm, are porphyritic (Ernst and Halls, 1984), but in Karelia they are principally not porphyritic. The Matachewan dykes are mainly Fe-rich quartz tholeiites, but most of the Karelian dykes are gabbronoritic–tholeiitic/Fe-tholeiitic in composition. Different compositions may be due to separate magma sources or due to different processes the dyke swarms experienced in Laurentia and the Fennoscandian Shield even if originating from the same source. Consequently, although the Matachewan and Karelian dykes are of the same age ca 2.45 Ga, paleomagnetic and compositional evidence cannot definitely confirm that the dykes were parts of the same swarm.

The above presented scenario is valid only, if the D component of the Karelian dykes is primary. Another candidate for the 2.44 Ga primary remanence is component D′ obtained from the Archean basement in Varisniemi. Here the primary nature of D′ for the remanence in dykes VN1 and VN7 is suggested from its occurrence in high unblocking temperatures (above component D), in baked Archean host rock and by the absence of the common superimposed secondary components A

and B. The reconstruction of Fig. 15(c) is compiled by using pole D′ (Table 4) and the pole from the Matachewan dykes (Bates and Halls, 1990). This reconstruction is closer to the one suggested by Heaman (1997), although Fennoscandia is rotated clockwise by ca 45°. In this reconstruction the Karelian and Matachewan dyke swarms are aligned. If the ca 1.0 Ga Grenvillian orogenic belt (Gower, 1996) between the two blocks is excluded, the Superior Province would have extended to the Kola Province in the SE and the Karelian dykes would have formed a parallel swarm to the Matachewan dykes. This reconstruction postulates a proximal origin for the dykes and thus the existence of a supercontinent, including Superior, Kola and Karelia. However, as noted before, pole D′ was obtained only from one site and therefore, much more data will be required to test it.

6. Conclusions

The 2.44 Ga mafic dykes in Russian Karelia carry hard and stable magnetizations, composed principally of three superimposed remanence components: A, B and D. The most common component A occurs in all dykes and in the Archean host rocks, except at one site (VN) in Varisniemi. Component A represents a partial remagnetization acquired during the Svecofennian orogeny at ca 1.9–1.84 Ga ago. Component A is typically isolated in a temperature range of ca 400–560°C. Another component, B, is also inferred to be a partial remagnetization isolated at about the same temperatures as component A, but always at lower temperatures when A and B occur within a same specimen. Component B is interpreted to have been acquired during the waning stage of the Svecofennian orogeny, ca 1.75 Ga ago. U–Pb datings on sphene and apatite (Amelin et al., 1995) of the age of 1.76 Ga from the nearby Olanga layered intrusions support this age interpretation. However, on the basis of the similarity of component B with remanence directions regarded as 2.1, 1.3 or 0.3 Ga elsewhere in the Fennoscandian shield, the age of component B is uncertain.

Component D is thought to represent the primary magnetization of the dykes, acquired ca 2.44 Ga ago. The primary nature of component D is supported by the following evidence:

1. D is isolated in narrow, high unblocking temperatures of 560–600°C, typical for remanent magnetization in basic igneous rocks (Butler, 1992).
2. Component D is best preserved in a petrographically fresh dyke (BD) which yields a U–Pb baddeleyite age of $2446+5/-4$ Ma (Vuollo et al., 1999a).
3. A similar remanence direction is typically isolated in ca 2.44 Ga old layered intrusions and related dykes all over in the Fennoscandian Shield.

However, baked contact tests for component D are negative suggesting that it is secondary. On the basis of the widespread occurrence of ca 2.44 Ga old K-granites, layered intrusions, volcanics and spatially dense mafic dykes in the area, it is proposed that the negative baked contact test is due to remagnetization of the Archean crust at about 2.44 Ga, at the same time as the emplacement of the mafic dykes.

Another candidate for the 2.44 Ga old magnetization is component D′ isolated from one site (VN) in Archean basement, in the vicinity of dykes VN1 and VN7 which are compositionally similar to dykes dated ca 2.44 Ga. The primary nature of D′ is supported by its occurrence in higher unblocking temperatures than component D, in an area that has escaped Svecofennian partial remagnetization.

If component D is regarded as the primary remanence of the dykes, then paleocontinental reconstruction of Laurentia and the Fennoscandian Shield places the Kola Peninsula adjacent to the southern margin of Greenland, thus spatially far from the Superior Province. If component D′ is regarded as a primary 2.44 Ga remanence, then the Fennoscandian Shield is located east of the Superior Province, more in alignment with the Matachewan dykes. However, acceptance of component D′ as the primary 2.44 Ga magnetization requires further studies on 2.44 Ga formations of the Fennoscandian Shield. Present indications for a general occurrence of component D′ are sparse.

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